

OPTILIFTS

Software Requirements Specification

COS 301 - Capstone Project | Demo 1
University of Pretoria | 2026

4.2 Design Specifications

OptiLifts follows a bold, athletic visual identity inspired by premium performance tools and modern sports analytics platforms. The design language prioritises clarity, data density, and a sense of intelligent power - reflecting the AI-driven nature of the system. Two official themes are provided: Dark Crimson (default) and Ivory & Crimson (light mode). Both themes use identical layout, typography, and component structure - only the colour values differ.

4.2.1 Brand Style Guide

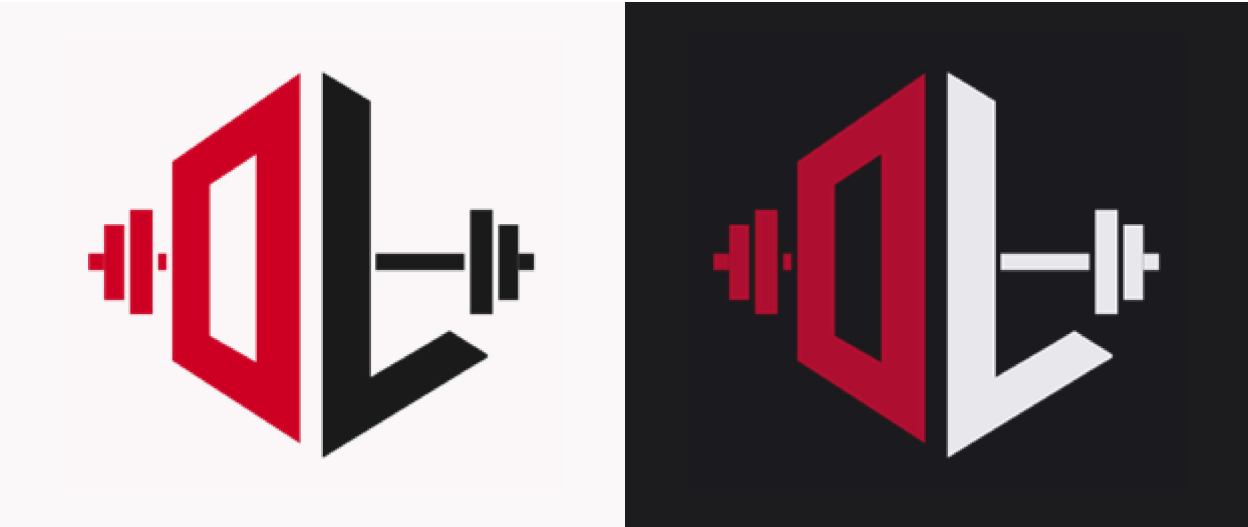
Logo

The OptiLifts logo system consists of two independently usable elements: an icon mark and a wordmark. Each element is provided in a light and a dark variant - four static assets in total, with animated equivalents of the icon mark for motion contexts.

Icon Mark

The icon mark is a bold dumbbell-and-hexagon composition. The hexagon communicates structure and strength; the dumbbell communicates the core promise of the product: performance tracking. Two structural colours are used — the crimson accent for the primary shape, and a neutral for the supporting elements — both of which shift between themes to maintain legibility against any background.

Variant	Structural	Accent	File
Light	#1A1A1A	#CC0022	logo-light.svg
Dark	#E8E8EC	#B01030	logo-dark.svg

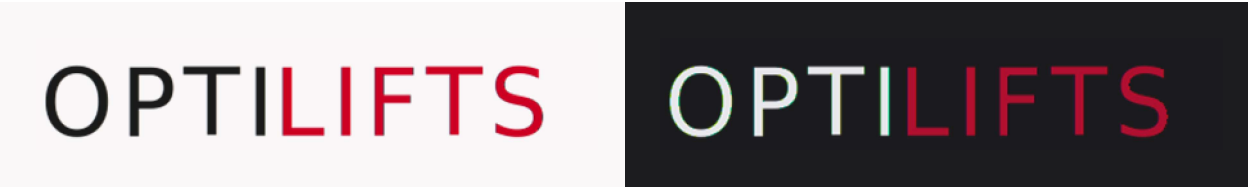


Two variants exist because the near-black structural colour becomes invisible against dark surfaces. The off-white variant ensures the mark reads cleanly on dark backgrounds without altering the composition.

Wordmark

Set in Bebas Neue, all caps, with wide letter-spacing. OPTI is rendered in the primary text colour and LIFTS in the accent colour, creating a deliberate visual split that reinforces the brand name. Both the text colour and accent shift between themes to maintain contrast.

Variant	OPTI	LIFTS	File
Light	#1A1A1A	#CC0022	wordmark-light.svg
Dark	#E8E8EC	#B01030	wordmark-dark.svg



Animated Variants

Animated versions of the icon mark are provided for splash screens, loading states, and web contexts where motion is appropriate. They follow the same two-variant rule and use identical colour values to their static counterparts. The animation starts without the weights in the logo. It then builds them piece by piece from the inner to outer parts. It does 5 benchpress reps and then deconstructs itself, looping to the beginning of the loop.

Variant	File
Light	logo-animated-light.svg
Dark	logo-animated-dark.svg

Logo Usage Rules

- Minimum wordmark height: 28px
- Always maintain the OPTI / LIFTS colour split - never render both words in the same colour
- Never stretch, rotate, or recolour the icon mark independently
- Minimum clear space: equal to the height of one capital letter on all sides
- Use the icon mark alone only when the OptiLifts name is already established in context

Colour Palette

OptiLifts ships with two official colour themes. All colours are specified in HEX.

Theme A - Dark Crimson (Default / Dark Mode)

Role	Name	HEX		Usage
Background	Charcoal Black		#1C1C1F	Page background
Surface	Elevated Surface		#26262B	Cards, sidebar, nav
Surface 2	Deep Surface		#2E2E34	Inputs, RPE buttons, chart bars
Border	Subtle Border		#3A3A42	Card borders, dividers, table lines
Accent Primary	Dark Crimson		#B01030	Buttons, active states, accent text, logo

Role	Name	HEX		Usage
Accent Secondary	Mid Crimson		#D94060	Progress fills, tags, secondary highlights
Primary Text	Off White		#E8E8EC	All body text, headings
Muted Text	Cool Grey		#9A9AA8	Labels, metadata, placeholders
Success	Muted Green		#4CAF50	Positive deltas, completion states
Warning	Amber		#FF9800	Fatigue warnings, negative deltas

Theme B - Ivory & Crimson (Light Mode)

Role	Name	HEX		Usage
Background	Warm White		#FAF8F8	Page background
Surface	Pure White		#FFFFFF	Cards, sidebar, nav
Surface 2	Blush Grey		#F5F0F0	Inputs, RPE buttons, chart bars
Border	Warm Border		#E8DEDE	Card borders, dividers, table lines
Accent Primary	Deep Crimson		#CC0022	Buttons, active states, accent text, logo
Accent Secondary	Dark Rose		#AA0018	Progress fills, tags, secondary highlights
Primary Text	Near Black		#1A1A1A	All body text, headings
Muted Text	Mid Grey		#666666	Labels, metadata, placeholders
Success	Forest Green		#1B6E1F	Positive deltas, completion states
Warning	Burnt Amber		#B35C00	Fatigue warnings, negative deltas

Typography

OptiLifts uses a two-font system: one display font for headings and a high-legibility font for body and UI text.

Role	Font	Weight	Size	Usage
Display / Headings	BEBAS NEUE	400	18–42px	Page titles, stat values, card titles, logo wordmark
Body / UI	Barlow	300/400/500/600	11–16px	Navigation, body copy, labels, buttons, metadata

Type scale:

Level	Font	Size	Weight	Letter Spacing
Page Title	Bebas Neue	42px	400	+2px
Section Title	Bebas Neue	18px	400	+1.5px
Card Value (stat)	Bebas Neue	40px	400	+1px
Button	Barlow	13px	700	+1px uppercase

Level	Font	Size	Weight	Letter Spacing
Body	Barlow	14px	400	0
Label / Meta	Barlow	10–12px	600–700	+1.5–2px uppercase

Rationale: Bebas Neue is a condensed display typeface widely associated with performance sports and athletic branding. Its tight vertical rhythm creates a strong visual hierarchy. Barlow provides excellent readability at small sizes and a wide weight range for UI hierarchy.

Iconography

OptiLifts uses simple line icons throughout the UI. Icons must:

- Be sourced from a consistent icon set (Lucide Icons)
- Always render at 18×18px in the sidebar and navigation
- Use `currentColor` so they inherit the parent element's colour automatically
- Have a minimum touch target of 44×44px on mobile per WCAG 2.5.5

Design Principles

- **1. Data first.**
 - Every screen must lead with the most important number or action. Stats are large, prominent, and always visible above the fold.
- **2. Progressive disclosure.**
 - The dashboard shows summary data. Drill-down reveals detail. No screen overwhelms the user with all available information at once.
- **3. AI is visible but not intrusive.**
 - AI insights appear in a clearly marked card. The AI never overrides the user's manual choices - it advises.
- **4. Consistency over creativity.**
 - All cards share the same border radius (0px), the same padding (22px), and the same header structure. No one-off styling.
- **5. Responsive by default.**
 - All layouts use Tailwind's responsive utility classes (`grid`, `flex`, `md:`, `lg:` prefixes). The sidebar uses `shadcn`'s sheet component on mobile viewports. Cards use Tailwind's responsive grid (`grid-cols-1 md:grid-cols-2 lg:grid-cols-4`) to reflow across breakpoints.
- **6. Accessibility is non-negotiable.**
 - All interactive elements have focus states. All colour combinations meet WCAG AA. All form inputs have associated labels.

UI Component Styling

Buttons

Variant	Background	Text	Border	Hover
Primary	<code>--accent</code>	White	None	Darken 10%
Ghost	Transparent	<code>--text</code>	<code>--border</code>	Border → accent, text → accent

All buttons: border-radius 4px · font-weight 700 · letter-spacing 1px · text-transform uppercase · font-size 13px

Cards

- Background: --surface
- Border: 1px solid --border
- Border radius: 10px
- Top accent on hover: 3px solid --accent
- Padding: 22px
- Header: 18px Bebas Neue, separated by a 1px --border divider

Navigation

- Height: 64px
- Active link: accent colour with 2px solid --accent bottom border
- Font: 13px Barlow 700 uppercase letter-spacing 1px

Stat Cards

- Four-column grid on desktop, two-column on tablet, single on mobile
- Value: 40px Bebas Neue --accent
- Label: 10px Barlow 700 uppercase --muted letter-spacing 2px
- Delta: 12px Barlow 600 in success/warning colour

Progress Bars

- Track: 4px height, --surface2 background
- Fill: --accent (primary) or --accent2 (secondary)
- End cap: 9×9px circular dot at fill endpoint

Responsive Breakpoints

Breakpoint	Layout Change
≥ 1200px	Header visible, 4-column stats grid, 2-column content grid
768px – 1199px	Header nav collapses to hamburger menu, 2-column stats grid
< 768px	Hamburger Sheet menu, single column layout, stats 2×2 grid

UI Component Styling

Buttons

Built on shadcn's Button component. Variants registered in components/ui/button.tsx via cva(). Tailwind overrides: rounded-sm font-bold tracking-wider uppercase text-xs.

Variant	shadcn variant	Tailwind classes	Usage
Primary	default	bg-accent text-white hover:bg-accent/90	Save, Start Session, Submit
Ghost	ghost	hover:border-accent hover:text-accent	Cancel, secondary actions

Cards

Built on shadcn's Card, CardHeader, CardContent components.

- Tailwind overrides: rounded-none border-border bg-surface p-[22px]
- Hover accent: hover:border-t-2 hover:border-t-accent transition-colors
- Header: 18px Bebas Neue, separated by a 1px border-border divider via CardHeader

Header Navigation

Built on shadcn's `NavigationMenu` (desktop) and `Sheet` (mobile). Active state managed via `Next.js usePathname()` hook.

- Height: `h-16` (64px) via Tailwind
- Active link: `border-b-2 border-accent text-accent`
- Font: `text-xs font-bold uppercase tracking-widest` (Barlow via CSS variable)
- Mobile: `Sheet` component triggered by `Button variant='ghost' size='icon'` with a Lucide Menu icon
- AI status indicator: small pulsing dot beside AI ACTIVE badge — `animate-pulse` via Tailwind

Stat Cards

Custom component built on shadcn's `Card`. Grid layout via Tailwind: `grid grid-cols-2 lg:grid-cols-4 gap-4`.

- Value: `text-[40px] font-display text-accent` (Bebas Neue via `font-display` CSS variable)
- Label: `text-[10px] font-bold uppercase tracking-[2px] text-muted`
- Delta: `text-xs font-semibold` in `text-success` or `text-warning`
- Full aria-label on each card reading the value, label, and delta as a sentence

Progress Bars

Built on shadcn's `Progress` component.

- Styled via Tailwind: `h-1 bg-surface-2 [&>div]:bg-accent`
- End-cap dot: absolutely positioned span with `w-2 h-2 rounded-full bg-accent` at fill endpoint
- `role='progressbar'` with `aria-valuenow`, `aria-valuemin`, `aria-valuemax` set automatically by shadcn

RPE Input

Custom component built on shadcn's `ToggleGroup` (`type='single'`).

- Grid layout: `grid grid-cols-5 gap-1`
- Selected state: `data-[state=on]:bg-accent data-[state=on]:text-white`
- Keyboard: `ToggleGroup` handles roving `tabindex` natively via Radix UI
- `aria-label='RPE rating, 1 to 10'` on the group; each `Toggle` has `aria-label='RPE [n]'`

Accessibility

OptiLifts targets WCAG 2.1 AA as the minimum standard across all screens and both themes. The implementation is split across four areas:

Colour Contrast

All primary text/background combinations meet WCAG 2.1 AA (4.5:1 minimum for normal text, 3:1 for large text and UI components):

Combination	Theme	Ratio	Standard
#E8E8EC on #1C1C1F	Dark	13.2:1	AAA
#B01030 on #1C1C1F	Dark	5.1:1	AA
#9A9AA8 on #1C1C1F	Dark	4.6:1	AA
#4CAF50 on #1C1C1F	Dark	5.8:1	AA
#FF9800 on #1C1C1F	Dark	8.2:1	AAA
#1A1A1A on #FAF8F8	Light	16.8:1	AAA

#CC0022 on #FFFFFF	Light	5.9:1	AA
#666666 on #FAF8F8	Light	5.2:1	AA

Keyboard Navigability

All functionality in OptiLifts must be operable via keyboard alone, with no mouse required:

- Tab order follows logical top-to-bottom, left-to-right reading flow on every screen
- shadcn Dialog (session summary, confirmation modals) traps focus inside the modal while open and returns focus to the trigger on close — handled automatically by Radix UI
- shadcn Sheet (mobile nav) follows the same focus trap behaviour
- shadcn ToggleGroup (RPE input, movement type selector) uses roving tabindex — arrow keys move between options, Enter/Space selects
- shadcn NavigationMenu supports full keyboard navigation including Escape to close submenus
- All custom components (muscle group chip selector, schedule day cells, exercise list rows) implement role, tabindex='0', and onKeyDown handlers for Enter and Space
- Active session set inputs are Tab-navigable in order: reps → weight → next set
- No keyboard trap exists anywhere in the application — users can always Tab out of any component
- All interactive elements show focus-visible:ring-2 focus-visible:ring-accent via Tailwind — never removed with outline:none without a replacement

Screen Reader Compatibility

OptiLifts is designed to be fully usable with screen readers. The following rules apply across the entire application:

- All form inputs use explicit labels via shadcn's FormLabel — placeholder text is never the sole label for an input
- Inline validation errors are linked to their input via aria-describedby, and the error element uses role='alert' so it is announced immediately on appearance
- Dynamic content that updates without a page navigation — such as counters, live feedback, and status changes — uses aria-live='polite' so updates are announced without interrupting the user
- Critical alerts that require immediate attention use aria-live='assertive'
- All progress indicators use role='progressbar' with aria-valuenow, aria-valuemin, and aria-valuemax so screen readers can report current progress as a value
- Interactive components that are not standard HTML elements expose the correct role, and their current state (selected, expanded, checked) via the appropriate aria-* attribute

WCAG 2.1 AA Compliance

Criterion	Description	Implementation
1.4.3	Contrast minimum (4.5:1 text)	All theme colour pairs verified — see table above
1.4.4	Resize text (200%)	Tailwind rem-based sizing; no fixed px text heights that break at zoom
1.4.11	Non-text contrast (3:1 for UI)	Accent buttons and focus rings meet 3:1 against adjacent surfaces
2.1.1	Keyboard operable	All shadcn/Radix primitives keyboard-native; custom components handle onKeyDown
2.4.3	Focus order	Tab order follows DOM order; no CSS-only reordering that breaks logical flow
2.4.7	Focus visible	focus-visible:ring-2 focus-visible:ring-accent on all interactive elements via Tailwind

3.3.1	Error identification	Inline errors via shadcn FormMessage; aria-describedby links error to input
3.3.2	Labels or instructions	All inputs have explicit labels via shadcn FormLabel; no placeholder-only labelling
4.1.2	Name, role, value	shadcn/Radix exposes correct ARIA roles; custom components assign role and aria-* manually
4.1.3	Status messages	Toast notifications and live region updates use aria-live='polite' or 'assertive'

4.2.2 Wireframes

The following wireframes represent the key screens for Demo 1. All screens are mid-fidelity - layout and component placement are finalised; final visual polish is applied in the live implementation.

Screen 1 - Register

Primary landing screen after login. Gives the athlete an at-a-glance view of their current training status and tonight's session.


[LOGIN](#)
[REGISTER](#)

REGISTER

Username

Email Address

Password

Re-enter Password

REGISTER

Already have an account? [Login](#)

Key interactions:

- "Start Session" button navigates to active workout mode
- RPE grid is clickable - selection updates fatigue log
- Chart bars are hoverable, showing session volume tooltip
- AI insight card is expandable for full reasoning

Screen 2 - Login



LOGIN

A light gray rounded rectangle containing the login form. At the top, the word "Username" is followed by a white text input field. Below that, the word "Password" is followed by a white text input field with a small eye icon on the right. Under the password field is a link that says "Forgot Password?". At the bottom of the form is a gray button with the text "LOGIN" in white. Below the button is a link that says "Don't have an account? Register".

Allow the athlete to add a new exercise to the library with name, movement type, and target muscle groups.

Key interactions:

- Name field validates on blur - shows inline error if empty or duplicate
- Movement type is a toggle - only one can be selected
- Muscle groups are multi-select chips - at least one required
- Save button disabled until name and at least one muscle group selected
- On success: toast notification + redirect to exercise library

Screen 3 - Create Workout

Allow the athlete to build a named workout by adding exercises with sets, reps, and weight.

OPTLIFTS

DASHBOARD

WORKOUTS

SCHEDULE

PROGRESS

PROFILE

CREATE WORKOUT

Workout Name

SAVE WORKOUT

Bicep Curl (Dumbbell)

Biceps

SET

KG

REPS

W

20

20

X

1

X

X

+ Add Set

Bicep Curl (Dumbbell)

Biceps

SET

KG

REPS

W

20

20

X

1

X

X

Recommended

Refresh

Bicep curl

Biceps

+

Bicep curl

Biceps

+

Bicep curl

Biceps

+

Exercises

+ Create Exercise

All Muscles

All Equipment

Search

Bicep curl

Biceps

+

Bicep curl

Biceps

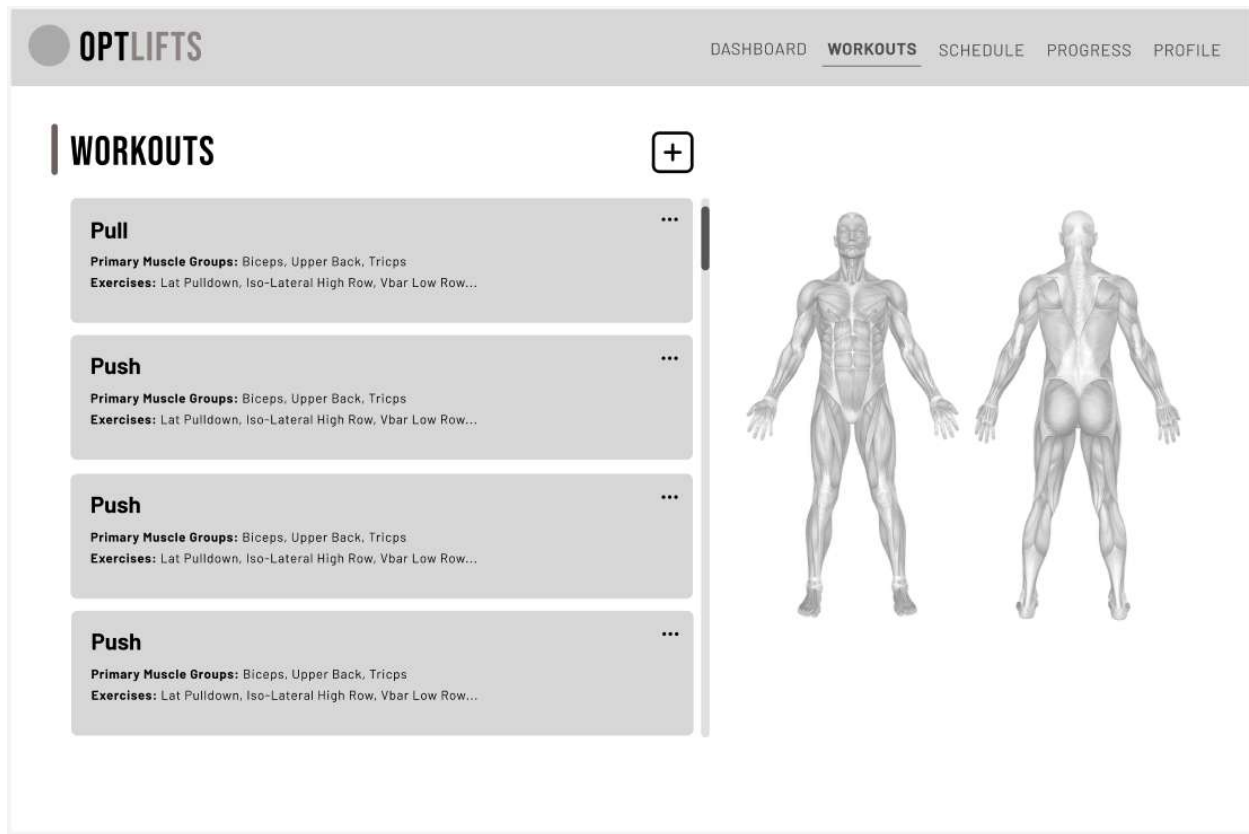
+

Key interactions:

- "Add Exercise" opens a search modal over the exercise library
- Each row has inline editable sets, reps, weight fields
- Rows are drag-reorderable
- Delete button removes the row with confirmation
- Save disabled until name + at least one exercise added

Screen 4 - Workout Page

Guide the athlete through their current session in real time - logging sets, weight, and RPE as they go.

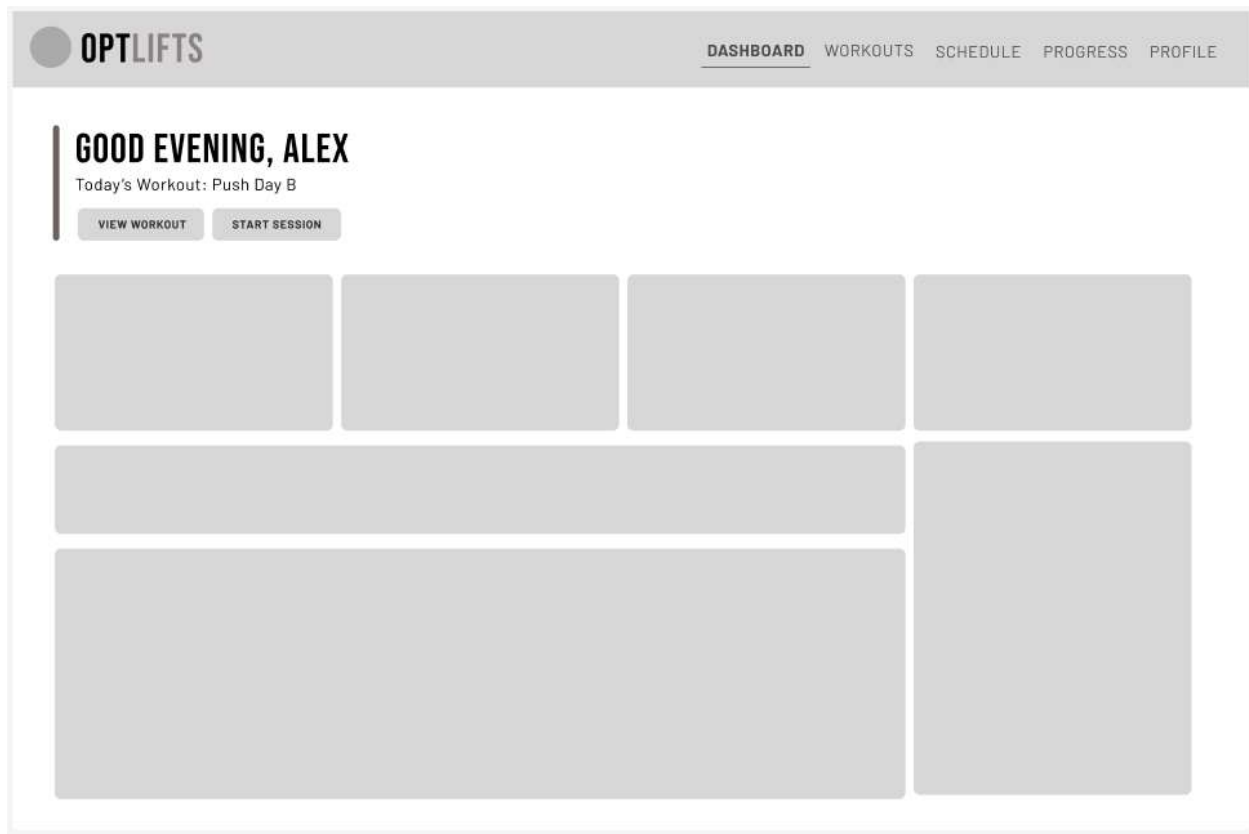


Key interactions:

- Sets are logged inline - reps and weight are editable
- Completed sets show a checkmark and become read-only
- Rest timer auto-starts after each completed set
- RPE logged per exercise, not per set
- "Finish Session" triggers summary + AI progression calculation

Screen 5 - Overview

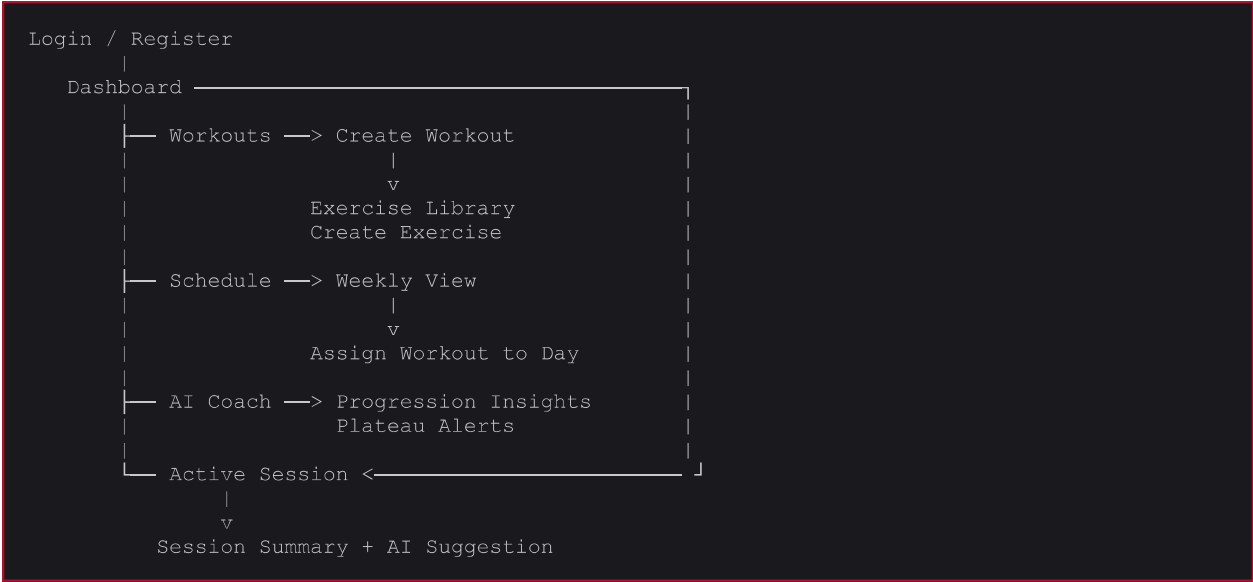
Show the user's full weekly workout schedule and allow them to assign workouts to days.



Key interactions:

- Day cells are clickable - opens workout assignment drawer
- Workouts can be dragged between days
- AI scheduling note appears when the system auto-adjusts
- Empty days show a + to add a workout
- Rest days are explicitly marked, not just empty

Navigation Flow



Theme Toggle

Implemented using next-themes (ThemeProvider wrapping the app). shadcn's built-in dark mode support uses the dark: Tailwind variant. Theme tokens are defined as CSS custom properties in globals.css under :root (light) and .dark (dark), mapped to Tailwind's theme.extend.colors in tailwind.config.ts. The toggle uses shadcn's Switch or DropdownMenu component.